

Improving Weight Perception in Virtual Reality via a Brain-Computer Interface

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ABSTRACT

The tracking offset between reality and virtual world helps users to generate pseudo-haptic perception of weight for virtual objects. However, the tracking offset mechanism is limited in its expressiveness in the sensation of weight since the tracking offset cannot be adjusted according to the user's actual perception of weight in real time. Here, a brain-computer interface (BCI) was used to detect the user's perception of the weight of virtual objects in real-time for adjustment of the tracking offset in a closed loop. We investigated the effect of this BCI mechanism via lifting ball game and compared it with other methods. The BCI mechanism not only has similar effect as the tracking offset mechanism but also significantly improves the consistency.

Index Terms: Weight perception, pseudo haptics, brain-computer interface, virtual reality.

1 INTRODUCTION

The weight perception in virtual world is based on the merging of multiple sources of sensory information (particularly the haptics and visual ones) [1, 2]. The pseudo-haptic feedback, in which the visual feedback simulates haptic sensations using the perception of object properties through visuo-haptic interaction, has been used to create weight perception of a virtual object [3,4]. The Offset mechanism is one of the solutions to help users perceive the weight of object in virtual reality (VR) [5]. An important concept in this mechanism is offset, which is defined as the vector between the virtual hand in VR and the tracked one in the real world. The offset has been proved to be helpful in generating pseudo-haptic weight perception. However, the Offset mechanism has its limitations in consistency and immersion. The existence of deviation causes users to mismatch real actions in VR. Furthermore, the sensitivity to offset is different for different users which would result in inconsistent feelings.

We propose a brain-computer interface (BCI) mechanism for real-time adjustment of the tracking offset according to the users' perception to improve user experience and keep consistency of perceiving weight. Compared to the stable offset during lifting virtual objects under the Offset mechanism, in our method, the tracking offset can be dynamically adjusted by user weight perception in real-time.

2 BCI BASED INTERACTION MECHANISM

In the Offset mechanism [5], the movement of virtual object can be described by the force model which contains two kinds of forces: the gravity of virtual objects and the imaginary force

generated by offset. The imaginary force from offset is against to the gravity. The value of imaginary force is correlated with the distance of offset. When the imaginary force is less than the gravity, the object would fall down with offset increase. On the contrary, when the imaginary force is larger than the gravity, the imaginary force will drag the object to close the position of tracked hand in the real world. In our modified model, there are three different forces. Besides the gravity and imaginary force correlated to offset, the third force is the imaginary force related to the feedback from BCI (Fig. 1). The direction of this force is coaxial with gravity. When the users' weight perception is lighter than the expected during lifting, f_{BCI} will help the gravity to pull down the object in which the offset is enlarged to enhance the weight perception. In the other side, the f_{BCI} is against the gravity, resulting in lower offset. Virtual object keeps still in space when the sum of three forces is zero, meaning the perceived weight of user matches to the expected one.

3 EXPERIMENTS

3.1 Participants

Twenty-two participants (8 females) aged 19 to 26 were recruited in this study. There were two experiments and each was involved 11 participants. All participants were equipped with HTC VIVE HMD, 32-channel Quik-Cap (Compumedics Neuroscan USA, Ltd. Charlotte, NC, USA) and Hi5 Glove (Fig. 1). With the Hi5 Glove, participants could control the virtual hand through their physical hand in VR.

At first, we trained a BCI model for each participant. In each trial during training, participants were asked to imagine lifting light ball or heavy ball according to the cue in a scene with default skybox. The EEG signals were collected when executing the imagine task. A training data of 40 trials (20 trials for imagining lifting light ball and 20 trials for imagining lifting heavy ball) was collected for each participant to train an SVM classifier for user weight perception identification. Before experiment, the participants were required to perform repeated training unless obtained the satisfactory accuracy of up to 70%.

In addition, for the 11 participants in the first experiment, a pre-experiment was performed to determine the best offset for three balls.

3.2 Experiment 1

In the experiment of ball lifting game, participants sat on a chair in front of a table in a room. During experiment, one of three balls: tennis, billiards, bowling, appeared in front of participants. Participants were required to lift the virtual ball and kept holding the ball with right hand to raise flat for a while. Then they could drop the ball down when a drawer closed and finished one trial. Participants were asked to report the weight they felt in this trial. Each ball appeared four times and the order was random. In this experiment, there were four conditions for weight perception. In condition 1, a ball with a bracket appeared in front of participants in the virtual scene. The participants were asked to estimate the weights of the virtual balls based on their initial impressions, without touching or lifting the balls. In condition 2, the

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participants were asked to lift up the ball without an offset and hold it for a while. In condition 3, we applied the *Offset* mechanism to the virtual ball. The offset for each kind of ball for each participant was that with the highest score in task 1. If two or more offsets had the highest score, we used their mean as the offset. In condition 4, we used the BCI mechanism for the virtual ball with an initial offset of 0 cm.

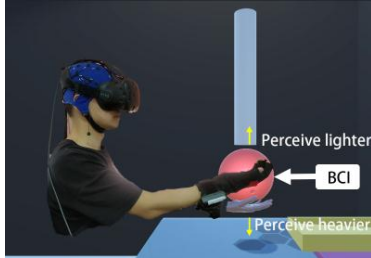


Figure 1. BCI mechanism. The offset was decreased through BCI when participant perceived lighter weight than expected, while the offset was increased when participant perceived heavier haptic weight.

3.3 Experiment 2

In this experiment, the aim is to examine the generalizability of the BCI and *Offset* mechanisms to new users without acceptable offset calibration for weight perception. Here, another 11 participants were recruited. The same virtual scene and virtual balls used in this task. However, when applying the *Offset* mechanism to a virtual ball, the offset for each kind of ball for each participant was that with the mean acceptable score from task 1. However, the initial offset was still 0 cm when the BCI mechanism was applied to the virtual ball. During each trial, the participants were asked to lift each of the three balls and determine its weight. There were 10 evaluation trials in total, 5 trials with balls applied with the *Offset* mechanism and 5 with the BCI mechanism. The participants evaluated all 5 trials with the same mechanism and then repeated the evaluation for the other 5 trials. The order of the evaluated mechanism for each participant was randomized and balanced.

4 RESULT

In the first experiment, all the mechanisms were presented for each participant, and the results were shown in the Fig. 2. Compared to the None condition, BCI and Offset mechanism can provide more realistic haptic weight perception. However, there was no significant difference between the BCI and Offset condition ($p=0.92$, Fig. 2).

In the second experiment, there was a significant difference between Offset and BCI condition ($p<0.05$, Fig. 3). The reported weight in the BCI condition was closer to the reality.

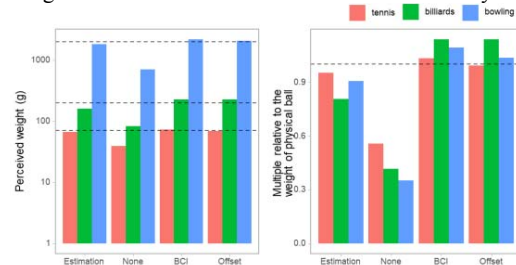


Figure 2. Left: the evaluated weight of tennis, billiards and bowling in four conditions. There was no significantly difference between the Offset and BCI mechanisms. The dashed lines represent the real weight of corresponding physical balls. Right: Boxplots of the ratios of the virtual weights to the real weights of the physical balls.

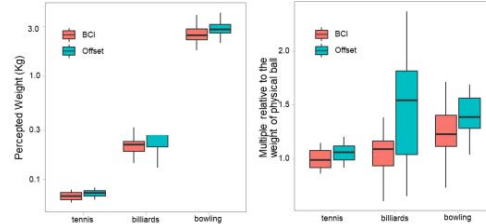


Figure 3. Results of experiment 2. Left: the perceived weight under the BCI and Offset conditions. Right: the multiple weight that is relative to the weight of physical ball of BCI and Offset condition.

5 CONCLUSION

In this poster, we proposed and evaluated BCI technique-based interaction mechanism to help users generate pseudo-haptic weight perception in VR. In our proposed approach, the offset between the virtual hand and the tracked one in reality can be dynamically adjusted by the BCI detection, which depends on the participants' weight perception of the virtual object according to the current tracking offset. If the tracking offset was too short to help the participant feel the expected weight of virtual objects, BCI would enlarge the offset until the pseudo-haptic weight perception was matched. On the contrary, the offset would be decreased for a lighter perception. In the first experiment, the BCI mechanism provided an effective simulation of weight as the Offset mechanism did. Furthermore, in the second experiment, the BCI could help the new participants perceive weight closer to the expected one compared to the Offset mechanism. The proposed mechanism didn't rely on the preset offset and has significant improvement in the consistency. Hence, we argue that our approach could help to address the weight aspect of haptics in all current tracking-based VR applications.

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